Reg. No. | | | | | | | | | | |

[14ME62]

SIXTH SEMESTER B.TECH EXAMINATION, JUNE – 2019

HEAT AND MASS TRANSFER

Time: 3 Hours

Maximum Marks:70

Instructions:

1. Answer any **FIVE** full questions, choosing at least **TWO** questions from each part.
2. Missing data, if any, may be suitably assumed.
3. Use of heat transfer data hand book permitted.

PART - A

1. a. Derive the three dimensional conduction equation in rectangular co-ordinate system **08**

   b. A reactor's wall, 320mm thick, is made up of an inner layer of fire brick (k = 0.84 W/m°C) covered with a layer of insulation (k = 0.16 W/m°C). The reactor operates at a temperature of 1325°C and ambient temperature is 25°C. Determine the thickness of the fire brick and the insulation which gives minimum heat loss. **04**

   c. State Fourier's law with expression. **02**

2. a. Explain efficiency and effectiveness of a fin. **04**

   b. State the assumptions made for the analysis of heat flow through a fin. **02**

   c. A turbine blade made of stainless steel (k = 29W/m°C) is 60mm long, 500mm^2 cross-sectional area and 120mm perimeter. The temperature of the root of the blade is 480°C and it is exposed to the products of combustion passing through the turbine at 820°C. If the film co-efficient between the blade and combustion gases is 320W/m^2°C, and the heat dissipation takes place from the blade by loosing heat at the tip, determine: **08**

   i. The temperature at the middle of the blade.

   ii. The rate of heat flow from the blade.

3. a. A steel ball 50mm in diameter and at 900°C is placed in still atmosphere of 30°C. Calculate the initial rate of cooling rate of the ball. Take $\rho = 7800 kg/m^3$, c=2kJ/kg°C, t = 1 min. **05**

   b. Prove that the temperature distribution in a body at time t during Newtonian heating or cooling is given by $\frac{T - T_\infty}{T_i - T_\infty} = e^{-B_i . F_o}$ **06**

   c. State the importance of Heisler Chart for transient heat conduction. **03**

4. a. Explain the concept of boundary layer on a flat plate. **06**

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b. Air is flowing over a flat plate 5m long and 2.5m wide with a velocity of 4m/s at 15°C if $\rho = 1.208 kg/m^3$ and V = 1.47 x 10^{-5} m^2/s. Calculate **08**

i. Length of the plate over which the boundary layer is laminar and the thickness of the boundary layer

ii. Shear stress at the location where the boundary layer exists.

iii. Total drag force on both sides of the plate.

PART-B

5. a. Explain the applications of dimensional analysis for free convection. **06**

b. A vertical plate measuring 180mmx180mm and at 50°C is exposed to atmosphere at 10°C. Compare the free convection heat transfer from this plate that would result due to forced convection over the plate at a velocity equal to twice the maximum velocity which would occur in the free convection boundary layer. **08**

6. a. Explain in brief the boiling regimes. **06**

b. Distinguish between filmwise condensation and dropwise condensation. **04**

c. Water is boiled at the rate of 25 kg/hr in a polished copper pan, 280mm in diameter at atmospheric pressure. Assuming nucleate boiling conditions, calculate the temperature at the bottom surface of the pan. **04**

7. a. Derive the expression for Logarithmic Mean Temperature Difference (LMTD) for counter flow heat exchangers. Also state its assumptions. **08**

b. The flow rates of hot and cold water streams running through a parallel flow heat exchanger are 0.2 kg/s and 0.5 kg/s respectively. The inlet temperatures on the hot and the cold sides are 75°C and 20°C respectively. The exit temperature of hot water is 45°C. If the individual heat transfer coefficients on both sides are 650 W/m^2°C, calculate the area of the heat exchanger. **06**

8. a Explain the concept of black body. **04**

b. State **04**

i. Wien's displacement Law.

ii. Stefan Boltzmann Law.

c. A surface having an area of 1.5m^2 and maintained at 300°C exchanges heat by radiation with another surface at 40°C. The value of factor due to the geometric location and emissivity is 0.52. Determine: **06**

i. Heat lost by radiation,

ii. The value of thermal resistance,

iii. The value of equivalent convection coefficient.